

Homework B3 | ECON 0100 | Fall 2026

Due: Sunday, Sep. 27

Homework is designed to both test your knowledge and challenge you to apply familiar concepts in new applications. Answer clearly and completely, and show your work so you can later understand your thought process. You are welcomed and encouraged to work in groups as long as your work is your own. Submit your answers on Gradescope when you're finished: each answer there is a selection.

Butterbeer

Preferences for butter beer can be represented by the following demand curve:

$$P_d = 100 - \frac{1}{2}Q_d$$

The supply curve for butter beer can be represented by the following equation:

$$P = 20 + \frac{1}{2}Q$$

Prices are in galleons and quantity is in bottles.

Q1 | Equilibrium

- a) Calculate equilibrium price: _____
- b) Find equilibrium quantity: _____
- c) Calculate consumer surplus: _____
- d) Calculate producer surplus: _____

Q2 | A Price Below Equilibrium

Butter beer again. Suppose the price is 50 galleons.

- a) What is the quantity demanded? _____
- b) What is the quantity supplied? _____
- c) Is this a shortage or an excess, and how large? _____
- d) Which way will the price move? _____

Q3 | A Price Above Equilibrium

Suppose instead the price is 70 galleons.

- a) What is the quantity demanded? _____
- b) What is the quantity supplied? _____
- c) Is this a shortage or an excess, and how large? _____
- d) Which way will the price move? _____